

KHOURDOUCH AMINE

Mechatronics, Robotics & Applied Physics Researcher

Automotive & Aerospace Mechatronics · Robotics & Cobotics · Mathematical & Theoretical Physics · Energy & Mechanical Engineering · Research & Innovation

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SCIENTIFIC PROFILE

Mechatronics researcher with an interdisciplinary background combining **robotics, mechanical engineering, applied physics, mathematical modelling, numerical simulation, embedded systems and artificial intelligence**. Currently pursuing a **Master d'Excellence in Mechatronics**, with a strong interest in research at the intersection of intelligent mechanical systems, autonomous robotics, and mathematical and applied physics.

My work combines **engineering design, theoretical analysis, mathematical modelling, computational simulation, and experimental investigation** to address complex scientific and technological challenges. My research interests include **quantum robotics, quantum machine learning, autonomous systems, nonlinear dynamics, multiphysics modelling, thermodynamics, energy systems, and Edge AI**.

EDUCATION

Master d'Excellence in Mechatronics

2025–2027

Centre d'Excellence Automobile et Aéronautique (CEAA), Université Ibn Tofaïl – Kénitra, Morocco

- Currently beginning the second year of the Master's program; expected graduation in July 2027.
- Coursework and projects covering mechatronics, robotics, intelligent systems, modelling, simulation, control, and multidisciplinary engineering.

Licence d'Excellence in Mechatronics

2024–2025

Centre d'Excellence Automobile et Aéronautique (CEAA), Université Ibn Tofaïl – Kénitra, Morocco

- Training in mechatronics, automation, robotics, electronics, mechanical engineering, and control systems.

DEUG in Physical Sciences – Physics

2020–2023

Université Moulay Ismaïl, Faculty of Sciences of Meknès – Morocco

- Academic training in fundamental physics, mathematics, scientific methods, and applied physical sciences.

Baccalaureate in Physical Sciences

2020

Lycée Qualifiant El Mansour Eddahbi – Morocco

RESEARCH INTERESTS

Robotics & Autonomous Systems

Autonomous Navigation & Path Planning

Nonlinear Dynamics & Dynamical Systems

Artificial Intelligence & Edge AI

Mechanical Engineering

Quantum Robotics & Quantum Machine Learning

Mathematical & Theoretical Physics

Applied Physics & Multiphysics

Thermodynamics & Energy Systems

Automotive & Aerospace Mechatronics

RESEARCH & R&D EXPERIENCE

Research Project – Thermoelectric Energy System

02/2024–07/2024

Laboratoire de Mécanique Énergétique – Faculty of Sciences of Meknès

- Conducted experimental and numerical investigation of a **Bi₂Te₃-based thermoelectric heat pump**.
- Analysed thermodynamic and thermoelectric behaviour under different electrical operating conditions.
- Modelled and simulated thermoelectric configurations using **COMSOL Multiphysics**.
- Performed experimental measurements using thermocouples and laboratory instrumentation.
- Analysed thermal performance and thermoelectric figure of merit.
- Result: 17/20 – First-ranked student.**

R&D Project – Additive Manufacturing & Materials Recycling

02/2025–06/2025

Espace Robotique & Factory – CEAA

- Developed a system for transforming waste from 3D printing into reusable polymer filament.
- Designed the mechanical architecture and integrated the electronic and thermal control systems.
- Used **CATIA, Proteus, CES EduPack, Python/C++, and Arduino**.
- Integrated **L298N, A4988, and MAX6675** modules for motor control and temperature measurement.
- Implemented thermal monitoring and process characterization.
- Experimentally obtained filament diameters in the range of **1.74–1.76 mm**.
- Result: 17/20 – First-ranked student.**

RESEARCH & ENGINEERING PROJECTS

Hybrid Quantum Machine Learning for Autonomous Robot Path Planning

Research Direction

Quantum Robotics – Simulation-Based Research

- Investigating hybrid quantum-classical approaches for autonomous robotic decision-making.
- Studying **Quantum Machine Learning, QAOA**, and optimization methods for robotic path planning.
- Exploring simulated quantum-sensing information for navigation under uncertainty.
- Comparing classical and quantum methodologies for robotic decision-making and optimization problems.

STM32-Based Predictive Maintenance with Edge AI

Embedded AI

- Designing an embedded predictive-maintenance platform based on **STM32**.
- Developing a multi-sensor architecture for local data acquisition and processing.
- Exploring **Edge AI** methods for anomaly and fault detection.
- Planned integration of **FreeRTOS, TensorFlow Lite Micro, and MQTT**.

ROS 2 / Gazebo Autonomous Robotic System

Robotics

- Developing and simulating autonomous robotic systems using **ROS 2 and Gazebo**.
- Studying robotic modelling, inter-node communication, simulation, and autonomous navigation.
- Working in a Linux-based environment with **ROS 2 Jazzy**.

Robotic Manipulation & Vision-Based Automation

Robotics / Computer Vision

- Studying robotic manipulation and vision-based automation systems.
- Applying computer-vision methods for robotic perception and decision-making.

Aerodynamic & Flight Simulation of a Sukhoi Aircraft

Aerospace Simulation

- Investigating aerodynamic behaviour and flight simulation of an aircraft.
- Applying numerical modelling methods to aerospace engineering problems.

Nonlinear Dynamics & Energy Transfer in Coupled Oscillatory Systems

Mathematical Physics

- Developing mathematical models of coupled oscillatory systems and nonlinear dynamics.
- Investigating energy-transfer mechanisms and collective dynamical behaviour.
- Exploring potential applications to resonant systems and wireless power transfer.

NUMERICAL SIMULATION & SCIENTIFIC COMPUTING

Forced-Convection Thermal Management of an Embedded System

Thermal / CFD

- Developed a thermal model of an embedded system based on **Raspberry Pi 4**.
- Simulated forced convection for airflow velocities ranging from **1–5 m/s**.
- Applied the **k- ω SST** turbulence model.
- Analysed the numerical thermal behaviour and interpreted simulation results.
- Simulation performed using **SimScale**.

Simulation Workflow:

Problem → Mathematical Model → Parameters → Numerical Method → Simulation → Results → Scientific Interpretation

TECHNICAL SKILLS

Robotics & Autonomous Systems: ROS 2, Gazebo, RoboDK, PX4, autonomous navigation, trajectory planning, robot simulation

Embedded Systems & IoT: STM32, ESP32, Raspberry Pi, Arduino, FreeRTOS, CAN Bus, Ethernet, MQTT, UART, SPI, I²C

Artificial Intelligence & Computer Vision: Python, OpenCV, TensorFlow, TensorFlow Lite Micro, PyTorch, Edge AI, object detection, image processing

Control & Automation: MATLAB/Simulink, PID, LQR, MPC, state estimation, dynamic systems, control modelling

Numerical & Multiphysics Simulation: COMSOL Multiphysics, ANSYS, SimScale, CFD, FEA, thermal modelling, multiphysics simulation

Mechanical Engineering & CAD: CATIA V5, SolidWorks, 3D modelling, assemblies, parametric design

Industrial Automation: Siemens TIA Portal, PLC, Ladder / LD, GRAFCET, industrial automation

Programming & Scientific Computing: Python, C, C++, Java, Linux, Bash, LaTeX, numerical programming

SCIENTIFIC WRITING & BOOK PROJECTS

Théorie moderne des métaux

Scientific Book Project

De Drude à Sommerfeld – Historical and Quantum Study of Electronic Conduction

Scientific book project focusing on the historical development and quantum foundations of electronic conduction, from classical Drude theory to Sommerfeld's quantum model.

LE MODÈLE STANDARD

Scientific Book Project

Théorie fondamentale de la matière et des interactions

Scientific book project addressing the theoretical framework of fundamental matter and its interactions.

FUTURE RESEARCH DIRECTIONS

Quantum Robotics Hybrid QML, QAOA, quantum sensing, path planning	Intelligent Autonomous Systems Autonomous navigation, perception, AI-based decision-making
Advanced Dynamical Systems Nonlinear dynamics, coupled oscillators, energy transfer	Multiphysics & Energy Systems Thermodynamics, heat transfer, intelligent energy systems

PUBLICATIONS & RESEARCH OUTPUT

Research publications currently in development.
Current research activities focus on simulation-based research, scientific modelling, computational methods, and the preparation of future peer-reviewed manuscripts, particularly in robotics, quantum machine learning, dynamical systems, and applied physics.

CERTIFICATES & PROFESSIONAL DEVELOPMENT

Robotics & ROS 2	Artificial Intelligence / Machine Learning
Embedded Systems	MATLAB / Simulink
COMSOL / Multiphysics Simulation	Python & Scientific Computing
Electronics & Control	Quantum Computing
Scientific English	Professional Engineering Development

LANGUAGES

Arabic: Native | **French:** Fluent / Excellent | **English:** B1 (Intermediate)

RESEARCH PROFILE

Engineering: Mechatronics, Robotics, Mechanics, Electronics, Control, Automotive, Aerospace
Physics: Thermodynamics, Applied Physics, Mathematical Physics, Theoretical Physics, Energy Systems
Computation: Scientific Computing, Numerical Modelling, Simulation, Artificial Intelligence, Machine Learning, Edge AI
Research: Mathematical Modelling, Multiphysics, Dynamical Systems, Quantum Robotics, Scientific Writing